

6

Numerical Solutions of Ordinary Differential Equations

EXERCISES 6.1

Euler Methods and Error Analysis

1. $h=0.1$

x_n	y_n
1.00	5.0000
1.10	3.9900
1.20	3.2546
1.30	2.7236
1.40	2.3451
1.50	2.0801

$h=0.05$

x_n	y_n
1.00	5.0000
1.05	4.4475
1.10	3.9763
1.15	3.5751
1.20	3.2342
1.25	2.9452
1.30	2.7009
1.35	2.4952
1.40	2.3226
1.45	2.1786
1.50	2.0592

2. $h=0.1$

x_n	y_n
0.00	2.0000
0.10	1.6600
0.20	1.4172
0.30	1.2541
0.40	1.1564
0.50	1.1122

$h=0.05$

x_n	y_n
0.00	2.0000
0.05	1.8150
0.10	1.6571
0.15	1.5237
0.20	1.4124
0.25	1.3212
0.30	1.2482
0.35	1.1916
0.40	1.1499
0.45	1.1217
0.50	1.1056

3. $h=0.1$

x_n	y_n
0.00	0.0000
0.10	0.1005
0.20	0.2030
0.30	0.3098
0.40	0.4234
0.50	0.5470

$h=0.05$

x_n	y_n
0.00	0.0000
0.05	0.0501
0.10	0.1004
0.15	0.1512
0.20	0.2028
0.25	0.2554
0.30	0.3095
0.35	0.3652
0.40	0.4230
0.45	0.4832
0.50	0.5465

4. $h=0.1$

x_n	y_n
0.00	1.0000
0.10	1.1110
0.20	1.2515
0.30	1.4361
0.40	1.6880
0.50	2.0488

$h=0.05$

x_n	y_n
0.00	1.0000
0.05	1.0526
0.10	1.1113
0.15	1.1775
0.20	1.2526
0.25	1.3388
0.30	1.4387
0.35	1.5556
0.40	1.6939
0.45	1.8598
0.50	2.0619

6.1 Euler Methods and Error Analysis

5. $h=0.1$

x_n	y_n
0.00	0.0000
0.10	0.0952
0.20	0.1822
0.30	0.2622
0.40	0.3363
0.50	0.4053

$h=0.05$

x_n	y_n
0.00	0.0000
0.05	0.0488
0.10	0.0953
0.15	0.1397
0.20	0.1823
0.25	0.2231
0.30	0.2623
0.35	0.3001
0.40	0.3364
0.45	0.3715
0.50	0.4054

6. $h=0.1$

x_n	y_n
0.00	0.0000
0.10	0.0050
0.20	0.0200
0.30	0.0451
0.40	0.0805
0.50	0.1266

$h=0.05$

x_n	y_n
0.00	0.0000
0.05	0.0013
0.10	0.0050
0.15	0.0113
0.20	0.0200
0.25	0.0313
0.30	0.0451
0.35	0.0615
0.40	0.0805
0.45	0.1022
0.50	0.1266

7. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5215
0.20	0.5362
0.30	0.5449
0.40	0.5490
0.50	0.5503

$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5116
0.10	0.5214
0.15	0.5294
0.20	0.5359
0.25	0.5408
0.30	0.5444
0.35	0.5469
0.40	0.5484
0.45	0.5492
0.50	0.5495

8. $h=0.1$

x_n	y_n
0.00	1.0000
0.10	1.1079
0.20	1.2337
0.30	1.3806
0.40	1.5529
0.50	1.7557

$h=0.05$

x_n	y_n
0.00	1.0000
0.05	1.0519
0.10	1.1079
0.15	1.1684
0.20	1.2337
0.25	1.3043
0.30	1.3807
0.35	1.4634
0.40	1.5530
0.45	1.6503
0.50	1.7560

9. $h=0.1$

x_n	y_n
1.00	1.0000
1.10	1.0095
1.20	1.0404
1.30	1.0967
1.40	1.1866
1.50	1.3260

$h=0.05$

x_n	y_n
1.00	1.0000
1.05	1.0024
1.10	1.0100
1.15	1.0228
1.20	1.0414
1.25	1.0663
1.30	1.0984
1.35	1.1389
1.40	1.1895
1.45	1.2526
1.50	1.3315

10. $h=0.1$

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5498
0.30	0.5744
0.40	0.5986
0.50	0.6224

$h=0.05$

x_n	y_n
0.00	0.5000
0.05	0.5125
0.10	0.5250
0.15	0.5374
0.20	0.5498
0.25	0.5622
0.30	0.5744
0.35	0.5866
0.40	0.5987
0.45	0.6106
0.50	0.6224

6.1 Euler Methods and Error Analysis

11. To obtain the analytic solution use the substitution $u = x + y - 1$. The resulting differential equation in $u(x)$ will be separable.

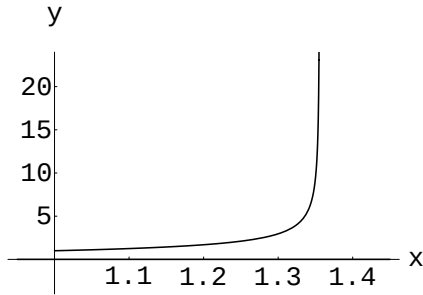
$h=0.1$

x_n	y_n	Actual Value
0.00	2.0000	2.0000
0.10	2.1220	2.1230
0.20	2.3049	2.3085
0.30	2.5858	2.5958
0.40	3.0378	3.0650
0.50	3.8254	3.9082

$h=0.05$

x_n	y_n	Actual Value
0.00	2.0000	2.0000
0.05	2.0553	2.1230
0.10	2.1228	2.3085
0.15	2.2056	2.5958
0.20	2.3075	3.0650
0.25	2.4342	3.9082
0.30	2.5931	2.5958
0.35	2.7953	2.7997
0.40	3.0574	3.0650
0.45	3.4057	3.4189
0.50	3.8840	3.9082

12. (a)



(b)

x_n	Euler	Imp. Euler
1.00	1.0000	1.0000
1.10	1.2000	1.2469
1.20	1.4938	1.6430
1.30	1.9711	2.4042
1.40	2.9060	4.5085

13. (a) Using Euler's method we obtain $y(0.1) \approx y_1 = 1.2$.

(b) Using $y'' = 4e^{2x}$ we see that the local truncation error is

$$y''(c) \frac{h^2}{2} = 4e^{2c} \frac{(0.1)^2}{2} = 0.02e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.02e^{0.2} = 0.0244$.

(c) Since $y(0.1) = e^{0.2} = 1.2214$, the actual error is $y(0.1) - y_1 = 0.0214$, which is less than 0.0244.

(d) Using Euler's method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.21$.

(e) The error in (d) is $1.2214 - 1.21 = 0.0114$. With global truncation error $O(h)$, when the step size is halved we expect the error for $h = 0.05$ to be one-half the error when $h = 0.1$. Comparing 0.0114 with 0.214 we see that this is the case.

14. (a) Using the improved Euler's method we obtain $y(0.1) \approx y_1 = 1.22$.

(b) Using $y''' = 8e^{2x}$ we see that the local truncation error is

$$y'''(c) \frac{h^3}{6} = 8e^{2c} \frac{(0.1)^3}{6} = 0.001333e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.001333e^{0.2} = 0.001628$.

(c) Since $y(0.1) = e^{0.2} = 1.221403$, the actual error is $y(0.1) - y_1 = 0.001403$ which is less than 0.001628.

(d) Using the improved Euler's method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.221025$.

6.1 Euler Methods and Error Analysis

- (e) The error in (d) is $1.221403 - 1.221025 = 0.000378$. With global truncation error $O(h^2)$, when the step size is halved we expect the error for $h = 0.05$ to be one-fourth the error for $h = 0.1$. Comparing 0.000378 with 0.001403 we see that this is the case.

15. (a) Using Euler's method we obtain $y(0.1) \approx y_1 = 0.8$.

- (b) Using $y'' = 5e^{-2x}$ we see that the local truncation error is

$$5e^{-2c} \frac{(0.1)^2}{2} = 0.025e^{-2c}.$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.025(1) = 0.025$.

- (c) Since $y(0.1) = 0.8234$, the actual error is $y(0.1) - y_1 = 0.0234$, which is less than 0.025.

- (d) Using Euler's method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.8125$.

- (e) The error in (d) is $0.8234 - 0.8125 = 0.0109$. With global truncation error $O(h)$, when the step size is halved we expect the error for $h = 0.05$ to be one-half the error when $h = 0.1$. Comparing 0.0109 with 0.0234 we see that this is the case.

16. (a) Using the improved Euler's method we obtain $y(0.1) \approx y_1 = 0.825$.

- (b) Using $y''' = -10e^{-2x}$ we see that the local truncation error is

$$10e^{-2c} \frac{(0.1)^3}{6} = 0.001667e^{-2c}.$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.001667(1) = 0.001667$.

- (c) Since $y(0.1) = 0.823413$, the actual error is $y(0.1) - y_1 = 0.001587$, which is less than 0.001667.

- (d) Using the improved Euler's method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.823781$.

- (e) The error in (d) is $|0.823413 - 0.8237181| = 0.000305$. With global truncation error $O(h^2)$, when the step size is halved we expect the error for $h = 0.05$ to be one-fourth the error when $h = 0.1$. Comparing 0.000305 with 0.001587 we see that this is the case.

17. (a) Using $y'' = 38e^{-3(x-1)}$ we see that the local truncation error is

$$y''(c) \frac{h^2}{2} = 38e^{-3(c-1)} \frac{h^2}{2} = 19h^2e^{-3(c-1)}.$$

- (b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$y''(c) \frac{h^2}{2} \leq 19(0.1)^2(1) = 0.19.$$

- (c) Using Euler's method with $h = 0.1$ we obtain $y(1.5) \approx 1.8207$. With $h = 0.05$ we obtain $y(1.5) \approx 1.9424$.

- (d) Since $y(1.5) = 2.0532$, the error for $h = 0.1$ is $E_{0.1} = 0.2325$, while the error for $h = 0.05$ is $E_{0.05} = 0.1109$. With global truncation error $O(h)$ we expect $E_{0.1}/E_{0.05} \approx 2$. We actually have $E_{0.1}/E_{0.05} = 2.10$.

18. (a) Using $y''' = -114e^{-3(x-1)}$ we see that the local truncation error is

$$\left| y'''(c) \frac{h^3}{6} \right| = 114e^{-3(c-1)} \frac{h^3}{6} = 19h^3e^{-3(c-1)}.$$

- (b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$\left| y'''(c) \frac{h^3}{6} \right| \leq 19(0.1)^3(1) = 0.019.$$

- (c) Using the improved Euler's method with $h = 0.1$ we obtain $y(1.5) \approx 2.080108$. With $h = 0.05$ we obtain $y(1.5) \approx 2.059166$.
- (d) Since $y(1.5) = 2.053216$, the error for $h = 0.1$ is $E_{0.1} = 0.026892$, while the error for $h = 0.05$ is $E_{0.05} = 0.005950$. With global truncation error $O(h^2)$ we expect $E_{0.1}/E_{0.05} \approx 4$. We actually have $E_{0.1}/E_{0.05} = 4.52$.
19. (a) Using $y'' = -1/(x+1)^2$ we see that the local truncation error is
- $$\left| y''(c) \frac{h^2}{2} \right| = \frac{1}{(c+1)^2} \frac{h^2}{2}.$$
- (b) Since $1/(x+1)^2$ is a decreasing function for $0 \leq x \leq 0.5$, $1/(c+1)^2 \leq 1/(0+1)^2 = 1$ for $0 \leq c \leq 0.5$ and
- $$\left| y''(c) \frac{h^2}{2} \right| \leq (1) \frac{(0.1)^2}{2} = 0.005.$$
- (c) Using Euler's method with $h = 0.1$ we obtain $y(0.5) \approx 0.4198$. With $h = 0.05$ we obtain $y(0.5) \approx 0.4124$.
- (d) Since $y(0.5) = 0.4055$, the error for $h = 0.1$ is $E_{0.1} = 0.0143$, while the error for $h = 0.05$ is $E_{0.05} = 0.0069$. With global truncation error $O(h)$ we expect $E_{0.1}/E_{0.05} \approx 2$. We actually have $E_{0.1}/E_{0.05} = 2.06$.
20. (a) Using $y''' = 2/(x+1)^3$ we see that the local truncation error is
- $$y'''(c) \frac{h^3}{6} = \frac{1}{(c+1)^3} \frac{h^3}{3}.$$
- (b) Since $1/(x+1)^3$ is a decreasing function for $0 \leq x \leq 0.5$, $1/(c+1)^3 \leq 1/(0+1)^3 = 1$ for $0 \leq c \leq 0.5$ and
- $$y'''(c) \frac{h^3}{6} \leq (1) \frac{(0.1)^3}{3} = 0.000333.$$
- (c) Using the improved Euler's method with $h = 0.1$ we obtain $y(0.5) \approx 0.405281$. With $h = 0.05$ we obtain $y(0.5) \approx 0.405419$.
- (d) Since $y(0.5) = 0.405465$, the error for $h = 0.1$ is $E_{0.1} = 0.000184$, while the error for $h = 0.05$ is $E_{0.05} = 0.000046$. With global truncation error $O(h^2)$ we expect $E_{0.1}/E_{0.05} \approx 4$. We actually have $E_{0.1}/E_{0.05} = 3.98$.
21. Because y_{n+1}^* depends on y_n and is used to determine y_{n+1} , all of the y_n^* cannot be computed at one time independently of the corresponding y_n values. For example, the computation of y_4^* involves the value of y_3 .

EXERCISES 6.2

Runge-Kutta Methods

1.

x_n	y_n	Actual Value
0.00	2.0000	2.0000
0.10	2.1230	2.1230
0.20	2.3085	2.3085
0.30	2.5958	2.5958
0.40	3.0649	3.0650
0.50	3.9078	3.9082

6.2 Runge-Kutta Methods

2. In this problem we use $h = 0.1$. Substituting $w_2 = \frac{3}{4}$ into the equations in (4) in the text, we obtain

$$w_1 = 1 - w_2 = \frac{1}{4}, \quad \alpha = \frac{1}{2w_2} = \frac{2}{3}, \quad \text{and} \quad \beta = \frac{1}{2w_2} = \frac{2}{3}.$$

The resulting second-order Runge-Kutta method is

$$y_{n+1} = y_n + h \left(\frac{1}{4}k_1 + \frac{3}{4}k_2 \right) = y_n + \frac{h}{4}(k_1 + 3k_2)$$

where

$$k_1 = f(x_n, y_n)$$

$$k_2 = f \left(x_n + \frac{2}{3}h, y_n + \frac{2}{3}hk_1 \right).$$

The table compares the values obtained using this second-order Runge-Kutta method with the values obtained using the improved Euler's method.

x_n	Second –Order Runge –Kutta	Improved Euler
0.00	2.0000	2.0000
0.10	2.1213	2.1220
0.20	2.3030	2.3049
0.30	2.5814	2.5858
0.40	3.0277	3.0378
0.50	3.8002	3.8254

3.

x_n	y_n
1.00	5.0000
1.10	3.9724
1.20	3.2284
1.30	2.6945
1.40	2.3163
1.50	2.0533

4.

x_n	y_n
0.00	2.0000
0.10	1.6562
0.20	1.4110
0.30	1.2465
0.40	1.1480
0.50	1.1037

5.

x_n	y_n
0.00	0.0000
0.10	0.1003
0.20	0.2027
0.30	0.3093
0.40	0.4228
0.50	0.5463

6.

x_n	y_n
0.00	1.0000
0.10	1.1115
0.20	1.2530
0.30	1.4397
0.40	1.6961
0.50	2.0670

7.

x_n	y_n
0.00	0.0000
0.10	0.0953
0.20	0.1823
0.30	0.2624
0.40	0.3365
0.50	0.4055

8.

x_n	y_n
0.00	0.0000
0.10	0.0050
0.20	0.0200
0.30	0.0451
0.40	0.0805
0.50	0.1266

9.

x_n	y_n
0.00	0.5000
0.10	0.5213
0.20	0.5358
0.30	0.5443
0.40	0.5482
0.50	0.5493

10.

x_n	y_n
0.00	1.0000
0.10	1.1079
0.20	1.2337
0.30	1.3807
0.40	1.5531
0.50	1.7561

11.

x_n	y_n
1.00	1.0000
1.10	1.0101
1.20	1.0417
1.30	1.0989
1.40	1.1905
1.50	1.3333

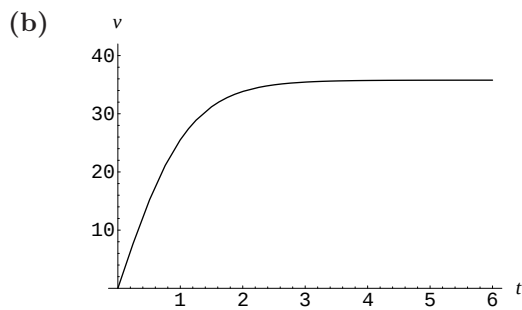
12.

x_n	y_n
0.00	0.5000
0.10	0.5250
0.20	0.5498
0.30	0.5744
0.40	0.5987
0.50	0.6225

13. (a) Write the equation in the form

$$\frac{dv}{dt} = 32 - 0.125v^2 = f(t, v).$$

t_n	v_n
0.0	0.0000
1.0	25.2570
2.0	32.9390
3.0	34.9770
4.0	35.5500
5.0	35.7130



- (c) Separating variables and using partial fractions we have

$$\frac{1}{2\sqrt{32}} \left(\frac{1}{\sqrt{32} - \sqrt{0.125}v} + \frac{1}{\sqrt{32} + \sqrt{0.125}v} \right) dv = dt$$

and

$$\frac{1}{2\sqrt{32}\sqrt{0.125}} \left(\ln |\sqrt{32} + \sqrt{0.125}v| - \ln |\sqrt{32} - \sqrt{0.125}v| \right) = t + c.$$

Since $v(0) = 0$ we find $c = 0$. Solving for v we obtain

$$v(t) = \frac{16\sqrt{5}(e^{\sqrt{3.2}t} - 1)}{e^{\sqrt{3.2}t} + 1}$$

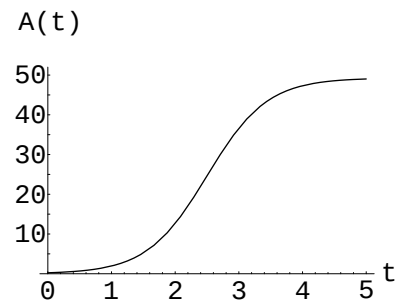
and $v(5) \approx 35.7678$. Alternatively, the solution can be expressed as

$$v(t) = \sqrt{\frac{mg}{k}} \tanh \sqrt{\frac{kg}{m}} t.$$

14. (a)

t (days)	1	2	3	4	5
A (observed)	2.78	13.53	36.30	47.50	49.40
A (approximated)	1.93	12.50	36.46	47.23	49.00

- (b) From the graph we estimate $A(1) \approx 1.68$, $A(2) \approx 13.2$, $A(3) \approx 36.8$, $A(4) \approx 46.9$, and $A(5) \approx 48.9$.



6.2 Runge-Kutta Methods

(c) Let $\alpha = 2.128$ and $\beta = 0.0432$. Separating variables we obtain

$$\begin{aligned}\frac{dA}{A(\alpha - \beta A)} &= dt \\ \frac{1}{\alpha} \left(\frac{1}{A} + \frac{\beta}{\alpha - \beta A} \right) dA &= dt \\ \frac{1}{\alpha} [\ln A - \ln(\alpha - \beta A)] &= t + c \\ \ln \frac{A}{\alpha - \beta A} &= \alpha(t + c) \\ \frac{A}{\alpha - \beta A} &= e^{\alpha(t+c)} \\ A &= \alpha e^{\alpha(t+c)} - \beta A e^{\alpha(t+c)} \\ \left[1 + \beta e^{\alpha(t+c)} \right] A &= \alpha e^{\alpha(t+c)}.\end{aligned}$$

Thus

$$A(t) = \frac{\alpha e^{\alpha(t+c)}}{1 + \beta e^{\alpha(t+c)}} = \frac{\alpha}{\beta + e^{-\alpha(t+c)}} = \frac{\alpha}{\beta + e^{-\alpha c} e^{-\alpha t}}.$$

From $A(0) = 0.24$ we obtain

$$0.24 = \frac{\alpha}{\beta + e^{-\alpha c}}$$

so that $e^{-\alpha c} = \alpha/0.24 - \beta \approx 8.8235$ and

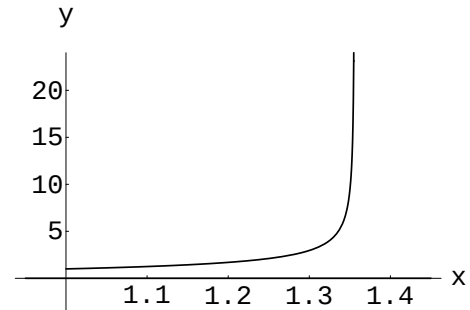
$$A(t) \approx \frac{2.128}{0.0432 + 8.8235e^{-2.128t}}.$$

t (days)	1	2	3	4	5
A (observed)	2.78	13.53	36.30	47.50	49.40
A (actual)	1.93	12.50	36.46	47.23	49.00

15. (a)

x_n	$h=0.05$	$h=0.1$
1.00	1.0000	1.0000
1.05	1.1112	
1.10	1.2511	1.2511
1.15	1.4348	
1.20	1.6934	1.6934
1.25	2.1047	
1.30	2.9560	2.9425
1.35	7.8981	
1.40	1.0608×10^{15}	903.0282

(b)



16. (a) Using the RK4 method we obtain $y(0.1) \approx y_1 = 1.2214$.

(b) Using $y^{(5)}(x) = 32e^{2x}$ we see that the local truncation error is

$$y^{(5)}(c) \frac{h^5}{120} = 32e^{2c} \frac{(0.1)^5}{120} = 0.000002667e^{2c}.$$

Since e^{2x} is an increasing function, $e^{2c} \leq e^{2(0.1)} = e^{0.2}$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.000002667e^{0.2} = 0.000003257$.

- (c) Since $y(0.1) = e^{0.2} = 1.221402758$, the actual error is $y(0.1) - y_1 = 0.000002758$ which is less than 0.000003257.
- (d) Using the RK4 formula with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 1.221402571$.
- (e) The error in (d) is $1.221402758 - 1.221402571 = 0.000000187$. With global truncation error $O(h^4)$, when the step size is halved we expect the error for $h = 0.05$ to be one-sixteenth the error for $h = 0.1$. Comparing 0.000000187 with 0.000002758 we see that this is the case.

17. (a) Using the RK4 method we obtain $y(0.1) \approx y_1 = 0.823416667$.

- (b) Using $y^{(5)}(x) = -40e^{-2x}$ we see that the local truncation error is

$$40e^{-2c} \frac{(0.1)^5}{120} = 0.000003333.$$

Since e^{-2x} is a decreasing function, $e^{-2c} \leq e^0 = 1$ for $0 \leq c \leq 0.1$. Thus an upper bound for the local truncation error is $0.000003333(1) = 0.000003333$.

- (c) Since $y(0.1) = 0.823413441$, the actual error is $|y(0.1) - y_1| = 0.000003225$, which is less than 0.000003333.
- (d) Using the RK4 method with $h = 0.05$ we obtain $y(0.1) \approx y_2 = 0.823413627$.
- (e) The error in (d) is $|0.823413441 - 0.823413627| = 0.000000185$. With global truncation error $O(h^4)$, when the step size is halved we expect the error for $h = 0.05$ to be one-sixteenth the error when $h = 0.1$. Comparing 0.000000185 with 0.000003225 we see that this is the case.

18. (a) Using $y^{(5)} = -1026e^{-3(x-1)}$ we see that the local truncation error is

$$\left| y^{(5)}(c) \frac{h^5}{120} \right| = 8.55h^5 e^{-3(c-1)}.$$

- (b) Since $e^{-3(x-1)}$ is a decreasing function for $1 \leq x \leq 1.5$, $e^{-3(c-1)} \leq e^{-3(1-1)} = 1$ for $1 \leq c \leq 1.5$ and

$$y^{(5)}(c) \frac{h^5}{120} \leq 8.55(0.1)^5(1) = 0.0000855.$$

- (c) Using the RK4 method with $h = 0.1$ we obtain $y(1.5) \approx 2.053338827$. With $h = 0.05$ we obtain $y(1.5) \approx 2.053222989$.

19. (a) Using $y^{(5)} = 24/(x+1)^5$ we see that the local truncation error is

$$y^{(5)}(c) \frac{h^5}{120} = \frac{1}{(c+1)^5} \frac{h^5}{5}.$$

- (b) Since $1/(x+1)^5$ is a decreasing function for $0 \leq x \leq 0.5$, $1/(c+1)^5 \leq 1/(0+1)^5 = 1$ for $0 \leq c \leq 0.5$ and

$$y^{(5)}(c) \frac{h^5}{5} \leq (1) \frac{(0.1)^5}{5} = 0.000002.$$

- (c) Using the RK4 method with $h = 0.1$ we obtain $y(0.5) \approx 0.405465168$. With $h = 0.05$ we obtain $y(0.5) \approx 0.405465111$.

20. Each step of Euler's method requires only 1 function evaluation, while each step of the improved Euler's method requires 2 function evaluations – once at (x_n, y_n) and again at (x_{n+1}, y_{n+1}^*) . The second-order Runge-Kutta methods require 2 function evaluations per step, while the RK4 method requires 4 function evaluations per step. To compare the methods we approximate the solution of $y' = (x + y - 1)^2$, $y(0) = 2$, at $x = 0.2$ using $h = 0.1$

6.2 Runge-Kutta Methods

for the Runge-Kutta method, $h = 0.05$ for the improved Euler's method, and $h = 0.025$ for Euler's method. For each method a total of 8 function evaluations is required. By comparing with the exact solution we see that the RK4 method appears to still give the most accurate result.

x_n	Euler $h=0.025$	Imp. Euler $h=0.05$	RK4 $h=0.1$	Actual
0.000	2.0000	2.0000	2.0000	2.0000
0.025	2.0250			2.0263
0.050	2.0526	2.0553		2.0554
0.075	2.0830			2.0875
0.100	2.1165	2.1228	2.1230	2.1230
0.125	2.1535			2.1624
0.150	2.1943	2.2056		2.2061
0.175	2.2395			2.2546
0.200	2.2895	2.3075	2.3085	2.3085

21. (a) For $y' + y = 10 \sin 3x$ an integrating factor is e^x so that

$$\begin{aligned} \frac{d}{dx}[e^x y] &= 10e^x \sin 3x \implies e^x y = e^x \sin 3x - 3e^x \cos 3x + c \\ \implies y &= \sin 3x - 3 \cos 3x + ce^{-x}. \end{aligned}$$

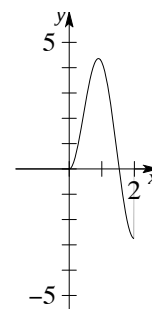
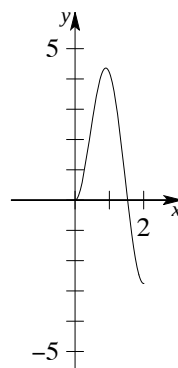
When $x = 0$, $y = 0$, so $0 = -3 + c$ and $c = 3$. The solution is

$$y = \sin 3x - 3 \cos 3x + 3e^{-x}.$$

Using Newton's method we find that $x = 1.53235$ is the only positive root in $[0, 2]$.

- (b) Using the RK4 method with $h = 0.1$ we obtain the table of values shown. These values are used to obtain an interpolating function in *Mathematica*. The graph of the interpolating function is shown. Using *Mathematica's* root finding capability we see that the only positive root in $[0, 2]$ is $x = 1.53236$.

x_n	y_n	x_n	y_n
0.0	0.0000	1.0	4.2147
0.1	0.1440	1.1	3.8033
0.2	0.5448	1.2	3.1513
0.3	1.1409	1.3	2.3076
0.4	1.8559	1.4	1.3390
0.5	2.6049	1.5	0.3243
0.6	3.3019	1.6	-0.6530
0.7	3.8675	1.7	-1.5117
0.8	4.2356	1.8	-2.1809
0.9	4.3593	1.9	-2.6061
1.0	4.2147	2.0	-2.7539



EXERCISES 6.3

Multistep Methods

In the tables in this section “ABM” stands for Adams-Bashforth-Moulton.

1. Writing the differential equation in the form $y' - y = x - 1$ we see that an integrating factor is $e^{-\int dx} = e^{-x}$, so that

$$\frac{d}{dx}[e^{-x}y] = (x-1)e^{-x}$$

and

$$y = e^x(-xe^{-x} + c) = -x + ce^x.$$

From $y(0) = 1$ we find $c = 1$, so the solution of the initial-value problem is $y = -x + e^x$. Actual values of the analytic solution above are compared with the approximated values in the table.

x_n	y_n	Actual	
0.0	1.00000000	1.00000000	init. cond.
0.2	1.02140000	1.02140276	RK4
0.4	1.09181796	1.09182470	RK4
0.6	1.22210646	1.22211880	RK4
0.8	1.42552788	1.42554093	ABM

2. The following program is written in *Mathematica*. It uses the Adams-Bashforth-Moulton method to approximate the solution of the initial-value problem $y' = x + y - 1$, $y(0) = 1$, on the interval $[0, 1]$.

```

Clear[f, x, y, h, a, b, y0];
f[x_, y_] := x + y - 1;          (* define the differential equation *)
h = 0.2;                          (* set the step size *)
a = 0; y0 = 1; b = 1;            (* set the initial condition and the interval *)
f[x, y]                          (* display the DE *)

Clear[k1, k2, k3, k4, x, y, u, v]
x = u[0] = a;
y = v[0] = y0;
n = 0;
While[x < a + 3h,                (* use RK4 to compute the first 3 values after y(0) *)
  n = n + 1;
  k1 = f[x, y];
  k2 = f[x + h/2, y + h k1/2];
  k3 = f[x + h/2, y + h k2/2];
  k4 = f[x + h, y + h k3];
  x = x + h;
  y = y + (h/6)(k1 + 2k2 + 2k3 + k4);
  u[n] = x;
  v[n] = y];

```

6.3 Multistep Methods

```

While[x ≤ b,                                     (* use Adams-Bashforth-Moulton *)
  p3 = f[u[n - 3], v[n - 3]];
  p2 = f[u[n - 2], v[n - 2]];
  p1 = f[u[n - 1], v[n - 1]];
  p0 = f[u[n], v[n]];
  pred = y + (h/24)(55p0 - 59p1 + 37p2 - 9p3);      (* predictor *)
  x = x + h;
  p4 = f[x, pred];
  y = y + (h/24)(9p4 + 19p0 - 5p1 + p2);          (* corrector *)
  n = n + 1;
  u[n] = x;
  v[n] = y]

(*display the table *)
TableForm[Prepend[Table[{u[n], v[n]}, {n, 0, (b-a)/h}], {"x(n)", "y(n)"}]];

```

3. The first predictor is $y_4^* = 0.73318477$.

x_n	y_n	
0.0	1.00000000	init. cond.
0.2	0.73280000	RK4
0.4	0.64608032	RK4
0.6	0.65851653	RK4
0.8	0.72319464	ABM

4. The first predictor is $y_4^* = 1.21092217$.

x_n	y_n	
0.0	2.00000000	init. cond.
0.2	1.41120000	RK4
0.4	1.14830848	RK4
0.6	1.10390600	RK4
0.8	1.20486982	ABM

5. The first predictor for $h = 0.2$ is $y_4^* = 1.02343488$.

x_n	h=0.2		h=0.1	
0.0	0.00000000	init. cond.	0.00000000	init. cond.
0.1			0.10033459	RK4
0.2	0.20270741	RK4	0.20270988	RK4
0.3			0.30933604	RK4
0.4	0.42278899	RK4	0.42279808	ABM
0.5			0.54631491	ABM
0.6	0.68413340	RK4	0.68416105	ABM
0.7			0.84233188	ABM
0.8	1.02969040	ABM	1.02971420	ABM
0.9			1.26028800	ABM
1.0	1.55685960	ABM	1.55762558	ABM

6. The first predictor for $h = 0.2$ is $y_4^* = 3.34828434$.

x_n	h=0.2		h=0.1	
0.0	1.00000000	init. cond.	1.00000000	init. cond.
0.1			1.21017082	RK4
0.2	1.44139950	RK4	1.44140511	RK4
0.3			1.69487942	RK4
0.4	1.97190167	RK4	1.97191536	ABM
0.5			2.27400341	ABM
0.6	2.60280694	RK4	2.60283209	ABM
0.7			2.96031780	ABM
0.8	3.34860927	ABM	3.34863769	ABM
0.9			3.77026548	ABM
1.0	4.22797875	ABM	4.22801028	ABM

7. The first predictor for $h = 0.2$ is $y_4^* = 0.13618654$.

x_n	h=0.2		h=0.1	
0.0	0.00000000	init. cond.	0.00000000	init. cond.
0.1			0.00033209	RK4
0.2	0.00262739	RK4	0.00262486	RK4
0.3			0.00868768	RK4
0.4	0.02005764	RK4	0.02004821	ABM
0.5			0.03787884	ABM
0.6	0.06296284	RK4	0.06294717	ABM
0.7			0.09563116	ABM
0.8	0.13598600	ABM	0.13596515	ABM
0.9			0.18370712	ABM
1.0	0.23854783	ABM	0.23841344	ABM

8. The first predictor for $h = 0.2$ is $y_4^* = 2.61796154$.

x_n	h=0.2		h=0.1	
0.0	1.00000000	init. cond.	1.00000000	init. cond.
0.1			1.10793839	RK4
0.2	1.23369623	RK4	1.23369772	RK4
0.3			1.38068454	RK4
0.4	1.55308554	RK4	1.55309381	ABM
0.5			1.75610064	ABM
0.6	1.99610329	RK4	1.99612995	ABM
0.7			2.28119129	ABM
0.8	2.62136177	ABM	2.62131818	ABM
0.9			3.02914333	ABM
1.0	3.52079042	ABM	3.52065536	ABM

EXERCISES 6.4

Higher-Order Equations and Systems

1. The substitution $y' = u$ leads to the iteration formulas

$$y_{n+1} = y_n + hu_n, \quad u_{n+1} = u_n + h(4u_n - 4y_n).$$

The initial conditions are $y_0 = -2$ and $u_0 = 1$. Then

$$y_1 = y_0 + 0.1u_0 = -2 + 0.1(1) = -1.9$$

$$u_1 = u_0 + 0.1(4u_0 - 4y_0) = 1 + 0.1(4 + 8) = 2.2$$

$$y_2 = y_1 + 0.1u_1 = -1.9 + 0.1(2.2) = -1.68.$$

The general solution of the differential equation is $y = c_1e^{2x} + c_2xe^{2x}$. From the initial conditions we find $c_1 = -2$ and $c_2 = 5$. Thus $y = -2e^{2x} + 5xe^{2x}$ and $y(0.2) \approx 1.4918$.

2. The substitution $y' = u$ leads to the iteration formulas

$$y_{n+1} = y_n + hu_n, \quad u_{n+1} = u_n + h\left(\frac{2}{x}u_n - \frac{2}{x^2}y_n\right).$$

The initial conditions are $y_0 = 4$ and $u_0 = 9$. Then

$$y_1 = y_0 + 0.1u_0 = 4 + 0.1(9) = 4.9$$

$$u_1 = u_0 + 0.1\left(\frac{2}{1}u_0 - \frac{2}{1}y_0\right) = 9 + 0.1[2(9) - 2(4)] = 10$$

$$y_2 = y_1 + 0.1u_1 = 4.9 + 0.1(10) = 5.9.$$

The general solution of the Cauchy-Euler differential equation is $y = c_1x + c_2x^2$. From the initial conditions we find $c_1 = -1$ and $c_2 = 5$. Thus $y = -x + 5x^2$ and $y(1.2) = 6$.

3. The substitution $y' = u$ leads to the system

$$y' = u, \quad u' = 4u - 4y.$$

Using formula (4) in the text with x corresponding to t , y corresponding to x , and u corresponding to y , we obtain the table shown.

x_n	$h=0.2$ y_n	$h=0.2$ u_n	$h=0.1$ y_n	$h=0.1$ u_n
0.0	-2.0000	1.0000	-2.0000	1.0000
0.1			-1.8321	2.4427
0.2	-1.4928	4.4731	-1.4919	4.4753

4. The substitution $y' = u$ leads to the system

$$y' = u, \quad u' = \frac{2}{x}u - \frac{2}{x^2}y.$$

Using formula (4) in the text with x corresponding to t , y corresponding to x , and u corresponding to y , we obtain the table shown.

x_n	$h=0.2$ y_n	$h=0.2$ u_n	$h=0.1$ y_n	$h=0.1$ u_n
1.0	4.0000	9.0000	4.0000	9.0000
1.1			4.9500	10.0000
1.2	6.0001	11.0002	6.0000	11.0000

6.4 Higher-Order Equations and Systems

5. The substitution $y' = u$ leads to the system

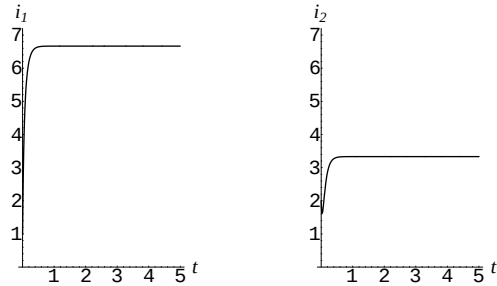
$$y' = u, \quad u' = 2u - 2y + e^t \cos t.$$

Using formula (4) in the text with y corresponding to x and u corresponding to y , we obtain the table shown.

x_n	$h=0.2$ y_n	$h=0.2$ u_n	$h=0.1$ y_n	$h=0.1$ u_n
0.0	1.0000	2.0000	1.0000	2.0000
0.1			1.2155	2.3150
0.2	1.4640	2.6594	1.4640	2.6594

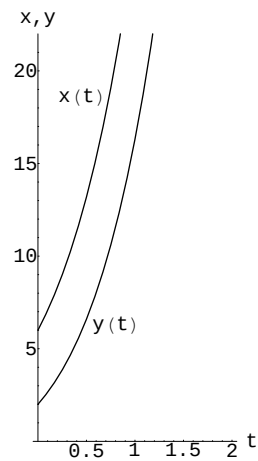
6. Using $h = 0.1$, the RK4 method for a system, and a numerical solver, we obtain

t_n	$h=0.2$ i_{1n}	$h=0.2$ i_{3n}
0.0	0.0000	0.0000
0.1	2.5000	3.7500
0.2	2.8125	5.7813
0.3	2.0703	7.4023
0.4	0.6104	9.1919
0.5	-1.5619	11.4877



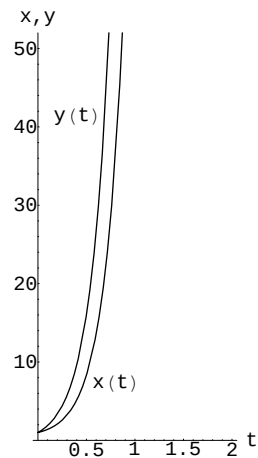
- 7.

t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	6.0000	2.0000	6.0000	2.0000
0.1			7.0731	2.6524
0.2	8.3055	3.4199	8.3055	3.4199



- 8.

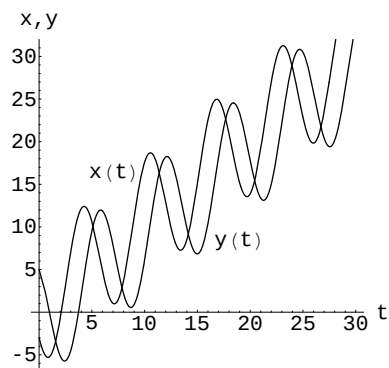
t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	1.0000	1.0000	1.0000	1.0000
0.1			1.4006	1.8963
0.2	2.0785	3.3382	2.0845	3.3502



6.4 Higher-Order Equations and Systems

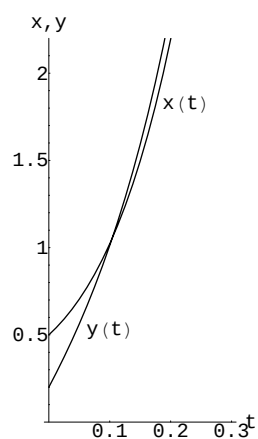
9.

t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	-3.0000	5.0000	-3.0000	5.0000
0.1			-3.4790	4.6707
0.2	-3.9123	4.2857	-3.9123	4.2857



10.

t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	0.5000	0.2000	0.5000	0.2000
0.1			1.0207	1.0115
0.2	2.1589	2.3279	2.1904	2.3592

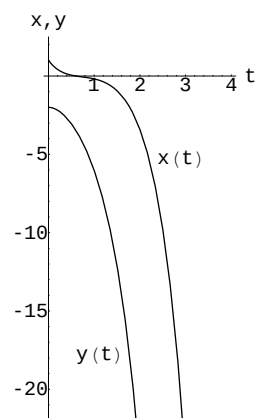


11. Solving for x' and y' we obtain the system

$$x' = -2x + y + 5t$$

$$y' = 2x + y - 2t.$$

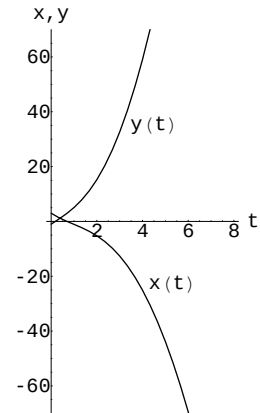
t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	1.0000	-2.0000	1.0000	-2.0000
0.1			0.6594	-2.0476
0.2	0.4179	-2.1824	0.4173	-2.1821



12. Solving for x' and y' we obtain the system

$$\begin{aligned}x' &= \frac{1}{2}y - 3t^2 + 2t - 5 \\y' &= -\frac{1}{2}y + 3t^2 + 2t + 5.\end{aligned}$$

t_n	$h=0.2$ x_n	$h=0.2$ y_n	$h=0.1$ x_n	$h=0.1$ y_n
0.0	3.0000	-1.0000	3.0000	-1.0000
0.1			2.4727	-0.4527
0.2	1.9867	0.0933	1.9867	0.0933



EXERCISES 6.5

Second-Order Boundary-Value Problems

1. We identify $P(x) = 0$, $Q(x) = 9$, $f(x) = 0$, and $h = (2 - 0)/4 = 0.5$. Then the finite difference equation is

$$y_{i+1} + 0.25y_i + y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	0.0	0.5	1.0	1.5	2.0
y	4.0000	-5.6774	-2.5807	6.3226	1.0000

2. We identify $P(x) = 0$, $Q(x) = -1$, $f(x) = x^2$, and $h = (1 - 0)/4 = 0.25$. Then the finite difference equation is

$$y_{i+1} - 2.0625y_i + y_{i-1} = 0.0625x_i^2.$$

The solution of the corresponding linear system gives

x	0.00	0.25	0.50	0.75	1.00
y	0.0000	-0.0172	-0.0316	-0.0324	0.0000

3. We identify $P(x) = 2$, $Q(x) = 1$, $f(x) = 5x$, and $h = (1 - 0)/5 = 0.2$. Then the finite difference equation is

$$1.2y_{i+1} - 1.96y_i + 0.8y_{i-1} = 0.04(5x_i).$$

The solution of the corresponding linear system gives

x	0.0	0.2	0.4	0.6	0.8	1.0
y	0.0000	-0.2259	-0.3356	-0.3308	-0.2167	0.0000

6.5 Second-Order Boundary-Value Problems

4. We identify $P(x) = -10$, $Q(x) = 25$, $f(x) = 1$, and $h = (1 - 0)/5 = 0.2$. Then the finite difference equation is

$$-y_i + 2y_{i-1} = 0.04.$$

The solution of the corresponding linear system gives

x	0.0	0.2	0.4	0.6	0.8	1.0
y	1.0000	1.9600	3.8800	7.7200	15.4000	0.0000

5. We identify $P(x) = -4$, $Q(x) = 4$, $f(x) = (1 + x)e^{2x}$, and $h = (1 - 0)/6 = 0.1667$. Then the finite difference equation is

$$0.6667y_{i+1} - 1.8889y_i + 1.3333y_{i-1} = 0.2778(1 + x_i)e^{2x_i}.$$

The solution of the corresponding linear system gives

x	0.0000	0.1667	0.3333	0.5000	0.6667	0.8333	1.0000
y	3.0000	3.3751	3.6306	3.6448	3.2355	2.1411	0.0000

6. We identify $P(x) = 5$, $Q(x) = 0$, $f(x) = 4\sqrt{x}$, and $h = (2 - 1)/6 = 0.1667$. Then the finite difference equation is

$$1.4167y_{i+1} - 2y_i + 0.5833y_{i-1} = 0.2778(4\sqrt{x_i}).$$

The solution of the corresponding linear system gives

x	1.0000	1.1667	1.3333	1.5000	1.6667	1.8333	2.0000
y	1.0000	-0.5918	-1.1626	-1.3070	-1.2704	-1.1541	-1.0000

7. We identify $P(x) = 3/x$, $Q(x) = 3/x^2$, $f(x) = 0$, and $h = (2 - 1)/8 = 0.125$. Then the finite difference equation is

$$\left(1 + \frac{0.1875}{x_i}\right)y_{i+1} + \left(-2 + \frac{0.0469}{x_i^2}\right)y_i + \left(1 - \frac{0.1875}{x_i}\right)y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	1.000	1.125	1.250	1.375	1.500	1.625	1.750	1.875	2.000
y	5.0000	3.8842	2.9640	2.2064	1.5826	1.0681	0.6430	0.2913	0.0000

8. We identify $P(x) = -1/x$, $Q(x) = x^{-2}$, $f(x) = \ln x/x^2$, and $h = (2 - 1)/8 = 0.125$. Then the finite difference equation is

$$\left(1 - \frac{0.0625}{x_i}\right)y_{i+1} + \left(-2 + \frac{0.0156}{x_i^2}\right)y_i + \left(1 + \frac{0.0625}{x_i}\right)y_{i-1} = 0.0156 \ln x_i.$$

The solution of the corresponding linear system gives

x	1.000	1.125	1.250	1.375	1.500	1.625	1.750	1.875	2.000
y	0.0000	-0.1988	-0.4168	-0.6510	-0.8992	-1.1594	-1.4304	-1.7109	-2.0000

9. We identify $P(x) = 1 - x$, $Q(x) = x$, $f(x) = x$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$[1 + 0.05(1 - x_i)]y_{i+1} + [-2 + 0.01x_i]y_i + [1 - 0.05(1 - x_i)]y_{i-1} = 0.01x_i.$$

The solution of the corresponding linear system gives

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	0.0000	0.2660	0.5097	0.7357	0.9471	1.1465	1.3353

0.7	0.8	0.9	1.0
1.5149	1.6855	1.8474	2.0000

10. We identify $P(x) = x$, $Q(x) = 1$, $f(x) = x$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$(1 + 0.05x_i)y_{i+1} - 1.99y_i + (1 - 0.05x_i)y_{i-1} = 0.01x_i.$$

The solution of the corresponding linear system gives

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	1.0000	0.8929	0.7789	0.6615	0.5440	0.4296	0.3216

0.7	0.8	0.9	1.0
0.2225	0.1347	0.0601	0.0000

11. We identify $P(x) = 0$, $Q(x) = -4$, $f(x) = 0$, and $h = (1 - 0)/8 = 0.125$. Then the finite difference equation is

$$y_{i+1} - 2.0625y_i + y_{i-1} = 0.$$

The solution of the corresponding linear system gives

x	0.000	0.125	0.250	0.375	0.500	0.625	0.750	0.875	1.000
y	0.0000	0.3492	0.7202	1.1363	1.6233	2.2118	2.9386	3.8490	5.0000

12. We identify $P(r) = 2/r$, $Q(r) = 0$, $f(r) = 0$, and $h = (4 - 1)/6 = 0.5$. Then the finite difference equation is

$$\left(1 + \frac{0.5}{r_i}\right) u_{i+1} - 2u_i + \left(1 - \frac{0.5}{r_i}\right) u_{i-1} = 0.$$

The solution of the corresponding linear system gives

r	1.0	1.5	2.0	2.5	3.0	3.5	4.0
u	50.0000	72.2222	83.3333	90.0000	94.4444	97.6190	100.0000

13. (a) The difference equation

$$\left(1 + \frac{h}{2}P_i\right) y_{i+1} + (-2 + h^2Q_i)y_i + \left(1 - \frac{h}{2}P_i\right) y_{i-1} = h^2f_i$$

is the same as equation (8) in the text. The equations are the same because the derivation was based only on the differential equation, not the boundary conditions. If we allow i to range from 0 to $n - 1$ we obtain n equations in the $n + 1$ unknowns y_{-1} , y_0 , y_1 , $\dots$, y_{n-1} . Since y_n is one of the given boundary conditions, it is not an unknown.

- (b) Identifying $y_0 = y(0)$, $y_{-1} = y(0 - h)$, and $y_1 = y(0 + h)$ we have from equation (5) in the text

$$\frac{1}{2h}[y_1 - y_{-1}] = y'(0) = 1 \quad \text{or} \quad y_1 - y_{-1} = 2h.$$

The difference equation corresponding to $i = 0$,

$$\left(1 + \frac{h}{2}P_0\right) y_1 + (-2 + h^2Q_0)y_0 + \left(1 - \frac{h}{2}P_0\right) y_{-1} = h^2f_0$$

becomes, with $y_{-1} = y_1 - 2h$,

$$\left(1 + \frac{h}{2}P_0\right) y_1 + (-2 + h^2Q_0)y_0 + \left(1 - \frac{h}{2}P_0\right) (y_1 - 2h) = h^2f_0$$

or

$$2y_1 + (-2 + h^2Q_0)y_0 = h^2f_0 + 2h - P_0.$$

Alternatively, we may simply add the equation $y_1 - y_{-1} = 2h$ to the list of n difference equations obtaining $n + 1$ equations in the $n + 1$ unknowns y_{-1} , y_0 , y_1 , $\dots$, y_{n-1} .

- (c) Using $n = 5$ we obtain

x	0.0	0.2	0.4	0.6	0.8	1.0
y	-2.2755	-2.0755	-1.8589	-1.6126	-1.3275	-1.0000

6.5 Second-Order Boundary-Value Problems

14. Using $h = 0.1$ and, after shooting a few times, $y'(0) = 0.43535$ we obtain the following table with the RK4 method.

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	1.00000	1.04561	1.09492	1.14714	1.20131	1.25633	1.31096

0.7	0.8	0.9	1.0
1.36392	1.41388	1.45962	1.50003

CHAPTER 6 REVIEW EXERCISES

1.

x_n	Euler h=0.1	Euler h=0.05	Imp. Euler h=0.1	Imp. Euler h=0.05	RK4 h=0.1	RK4 h=0.05
1.00	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000
1.05		2.0693		2.0735		2.0736
1.10	2.1386	2.1469	2.1549	2.1554	2.1556	2.1556
1.15		2.2328		2.2459		2.2462
1.20	2.3097	2.3272	2.3439	2.3450	2.3454	2.3454
1.25		2.4299		2.4527		2.4532
1.30	2.5136	2.5409	2.5672	2.5689	2.5695	2.5695
1.35		2.6604		2.6937		2.6944
1.40	2.7504	2.7883	2.8246	2.8269	2.8278	2.8278
1.45		2.9245		2.9686		2.9696
1.50	3.0201	3.0690	3.1157	3.1187	3.1197	3.1197

2.

x_n	Euler h=0.1	Euler h=0.05	Imp. Euler h=0.1	Imp. Euler h=0.05	RK4 h=0.1	RK4 h=0.05
0.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.05		0.0500		0.0501		0.0500
0.10	0.1000	0.1001	0.1005	0.1004	0.1003	0.1003
0.15		0.1506		0.1512		0.1511
0.20	0.2010	0.2017	0.2030	0.2027	0.2026	0.2026
0.25		0.2537		0.2552		0.2551
0.30	0.3049	0.3067	0.3092	0.3088	0.3087	0.3087
0.35		0.3610		0.3638		0.3637
0.40	0.4135	0.4167	0.4207	0.4202	0.4201	0.4201
0.45		0.4739		0.4782		0.4781
0.50	0.5279	0.5327	0.5382	0.5378	0.5376	0.5376

3.

x_n	Euler $h=0.1$	Euler $h=0.05$	Imp. Euler $h=0.1$	Imp. Euler $h=0.05$	RK4 $h=0.1$	RK4 $h=0.05$
0.50	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
0.55		0.5500		0.5512		0.5512
0.60	0.6000	0.6024	0.6048	0.6049	0.6049	0.6049
0.65		0.6573		0.6609		0.6610
0.70	0.7095	0.7144	0.7191	0.7193	0.7194	0.7194
0.75		0.7739		0.7800		0.7801
0.80	0.8283	0.8356	0.8427	0.8430	0.8431	0.8431
0.85		0.8996		0.9082		0.9083
0.90	0.9559	0.9657	0.9752	0.9755	0.9757	0.9757
0.95		1.0340		1.0451		1.0452
1.00	1.0921	1.1044	1.1163	1.1168	1.1169	1.1169

4.

x_n	Euler $h=0.1$	Euler $h=0.05$	Imp. Euler $h=0.1$	Imp. Euler $h=0.05$	RK4 $h=0.1$	RK4 $h=0.05$
1.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1.05		1.1000		1.1091		1.1095
1.10	1.2000	1.2183	1.2380	1.2405	1.2415	1.2415
1.15		1.3595		1.4010		1.4029
1.20	1.4760	1.5300	1.5910	1.6001	1.6036	1.6036
1.25		1.7389		1.8523		1.8586
1.30	1.8710	1.9988	2.1524	2.1799	2.1909	2.1911
1.35		2.3284		2.6197		2.6401
1.40	2.4643	2.7567	3.1458	3.2360	3.2745	3.2755
1.45		3.3296		4.1528		4.2363
1.50	3.4165	4.1253	5.2510	5.6404	5.8338	5.8446

5. Using

$$y_{n+1} = y_n + hu_n, \quad y_0 = 3$$

$$u_{n+1} = u_n + h(2x_n + 1)y_n, \quad u_0 = 1$$

we obtain (when $h = 0.2$) $y_1 = y(0.2) = y_0 + hu_0 = 3 + (0.2)1 = 3.2$. When $h = 0.1$ we have

$$y_1 = y_0 + 0.1u_0 = 3 + (0.1)1 = 3.1$$

$$u_1 = u_0 + 0.1(2x_0 + 1)y_0 = 1 + 0.1(1)3 = 1.3$$

$$y_2 = y_1 + 0.1u_1 = 3.1 + 0.1(1.3) = 3.23.$$

 6. The first predictor is $y_3^* = 1.14822731$.

x_n	y_n	
0.0	2.00000000	init. cond.
0.1	1.65620000	RK4
0.2	1.41097281	RK4
0.3	1.24645047	RK4
0.4	1.14796764	ABM

 7. Using $x_0 = 1$, $y_0 = 2$, and $h = 0.1$ we have

$$x_1 = x_0 + h(x_0 + y_0) = 1 + 0.1(1 + 2) = 1.3$$

$$y_1 = y_0 + h(x_0 - y_0) = 2 + 0.1(1 - 2) = 1.9$$

and

CHAPTER 6 REVIEW EXERCISES

$$x_2 = x_1 + h(x_1 + y_1) = 1.3 + 0.1(1.3 + 1.9) = 1.62$$

$$y_2 = y_1 + h(x_1 - y_1) = 1.9 + 0.1(1.3 - 1.9) = 1.84.$$

Thus, $x(0.2) \approx 1.62$ and $y(0.2) \approx 1.84$.

8. We identify $P(x) = 0$, $Q(x) = 6.55(1 + x)$, $f(x) = 1$, and $h = (1 - 0)/10 = 0.1$. Then the finite difference equation is

$$y_{i+1} + [-2 + 0.0655(1 + x_i)]y_i + y_{i-1} = 0.001$$

or

$$y_{i+1} + (0.0655x_i - 1.9345)y_i + y_{i-1} = 0.001.$$

The solution of the corresponding linear system gives

x	0.0	0.1	0.2	0.3	0.4	0.5	0.6
y	0.0000	4.1987	8.1049	11.3840	13.7038	14.7770	14.4083

0.7	0.8	0.9	1.0
12.5396	9.2847	4.9450	0.0000